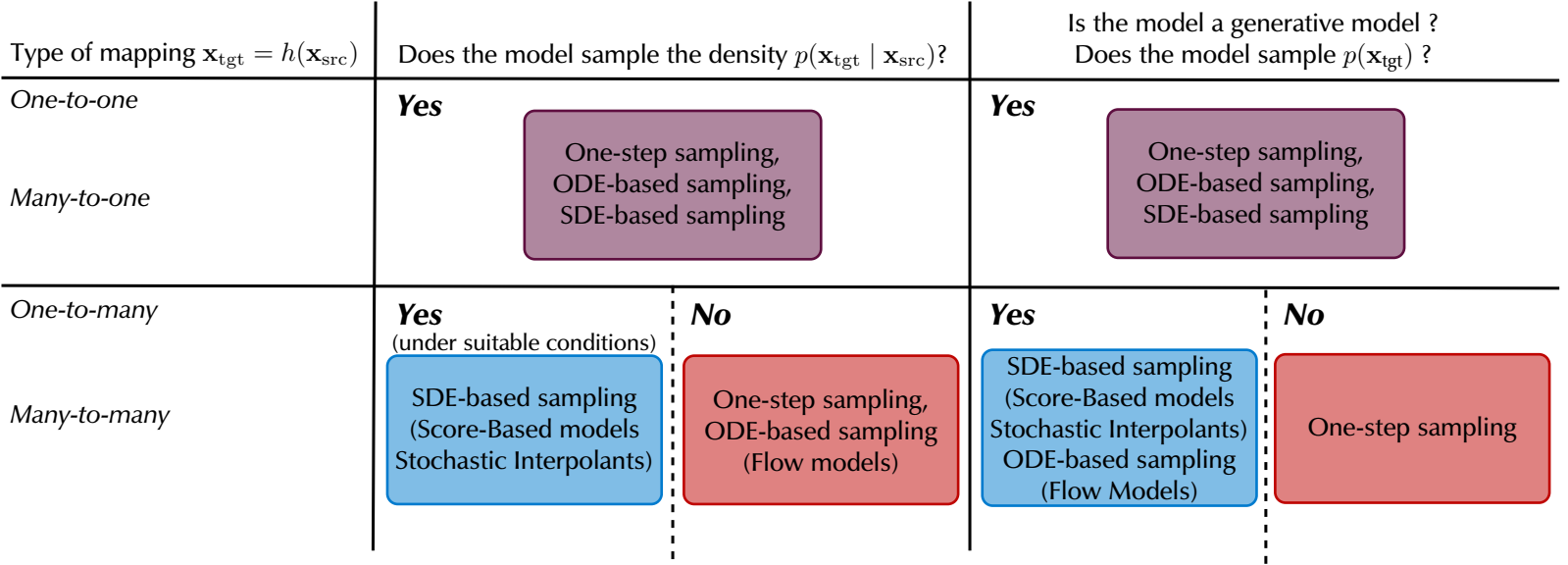

Type of mapping $\mathbf{x}_{\text{tgt}} = h(\mathbf{x}_{\text{src}})$
Does the model sample the density $p(\mathbf{x}_{\text{tgt}} \mid \mathbf{x}_{\text{src}})$?
Is the model a generative model ?
Does the model sample $p(\mathbf{x}_{\text{tgt}})$?
One-to-one
Many-to-one
Yes
One-step sampling,
ODE-based sampling,
SDE-based sampling
Yes
One-step sampling,
ODE-based sampling,
SDE-based sampling
One-to-many
Many-to-many
Yes
(under suitable conditions)
SDE-based sampling
(Score-Based models
Stochastic Interpolants)
No
One-step sampling,
ODE-based sampling
(Flow models)
Yes
SDE-based sampling
(Score-Based models
Stochastic Interpolants)
ODE-based sampling
(Flow Models)
No
One-step sampling